Summary: Natural vs. Orpheus PT base vs. Orpheus PT steered

cs-tts-steering project

November 5, 2026

Setup. Orpheus 3B Hindi *pretrained* (`canopylabs/3b-hi-pretrain-research_release`), text-only prompt (random speaker per clip), Devanagari/English transcripts, seed 1234. Steering: CAA vector (natural minus on-policy PT, 69 train pairs) added at layer 27 attention output, $\alpha = 1$. Groups: **Natural** (TV recordings), **PT base**, **PT steered**. Section 1 is the 19 held-out test utterances; Section 2 is all 100 utterances (in-sample). Full report: `preliminary_caa_pt.pdf`.

Headline.

- Test (19): no significant gain over PT base on M1–M3; steered beats a random vector on M1 (+0.79, $p = .023$). More failures (5 vs. 1), mostly duration flags.
- All 100 (in-sample): M1 +0.84 ($p = 5 \times 10^{-5}$), M2/M3 better ($p \approx .08$), syllable rate up, voice quality worse (HNR -1.1 dB). NISQA flat.
- Objective verdict: no-go, with a weak positive signal; listening pending.

Test set, medians	Natural	PT base	PT steered
WER (median)	.688	.750	.800
Median F0 (Hz)	220	129	149
F0 range (st)	19.9	6.3	9.3
HNR (dB)	8.7	15.5	15.0
NISQA-TTS	2.94	3.62	3.57
M1 logit (mean)	5.38	-4.25	-3.85

1 Test set (19 utterances)

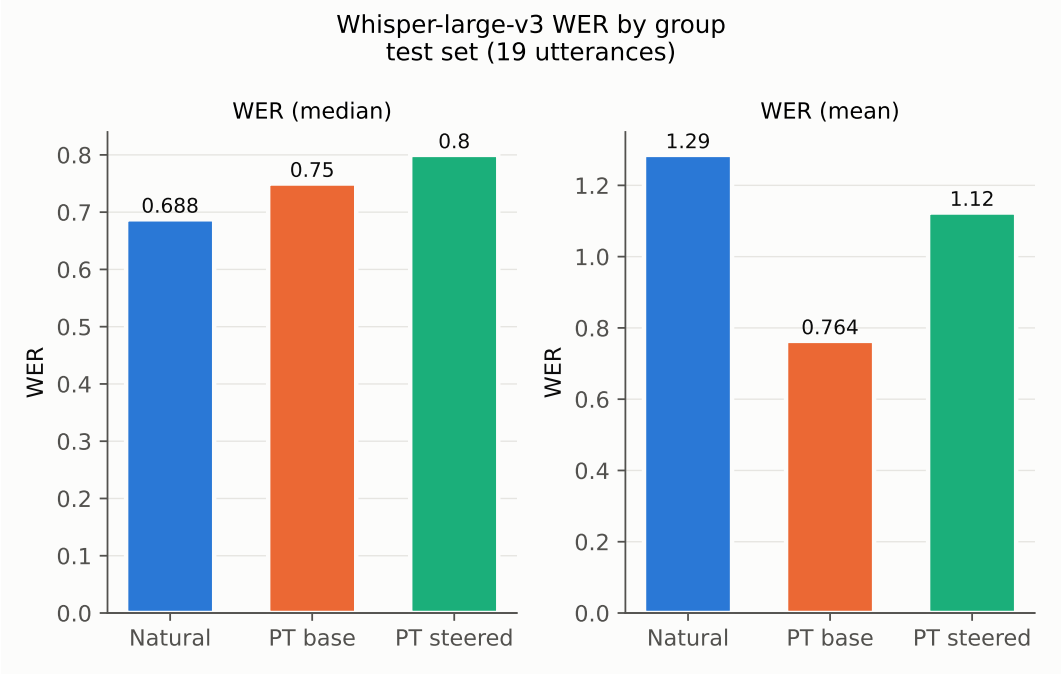


Figure 1: WER (Whisper-large-v3), median and mean. Natural WER is high, so WER is a weak intelligibility proxy here.

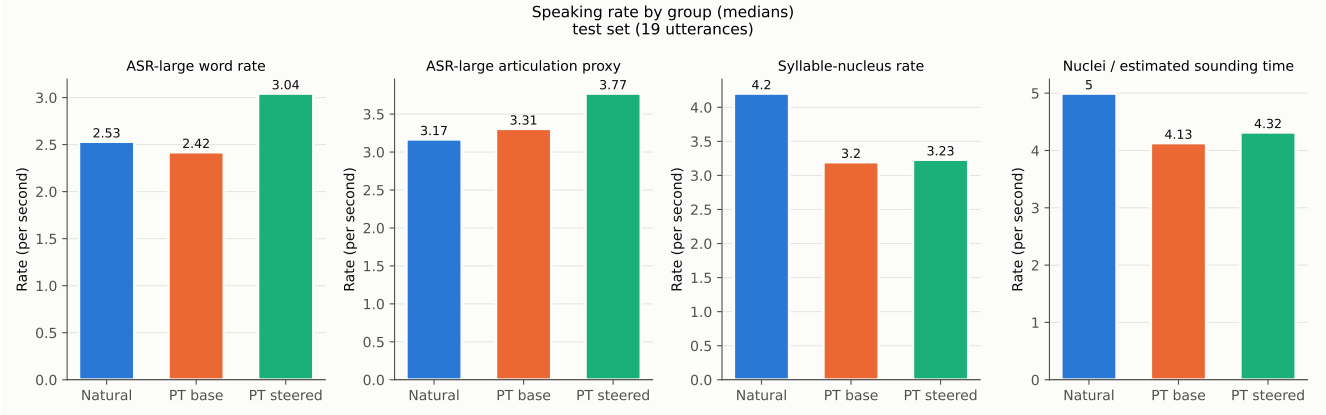


Figure 2: Speaking rate: ASR word rate, articulation proxy, syllable-nucleus rate, nuclei per sounding time.

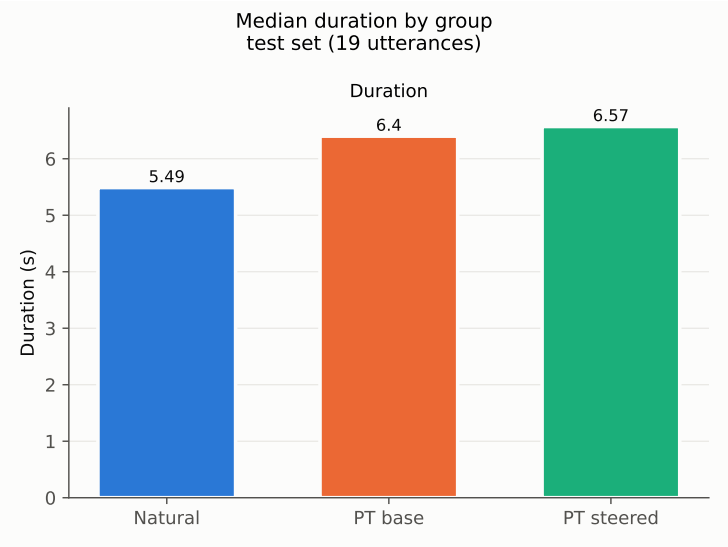


Figure 3: Median clip duration.

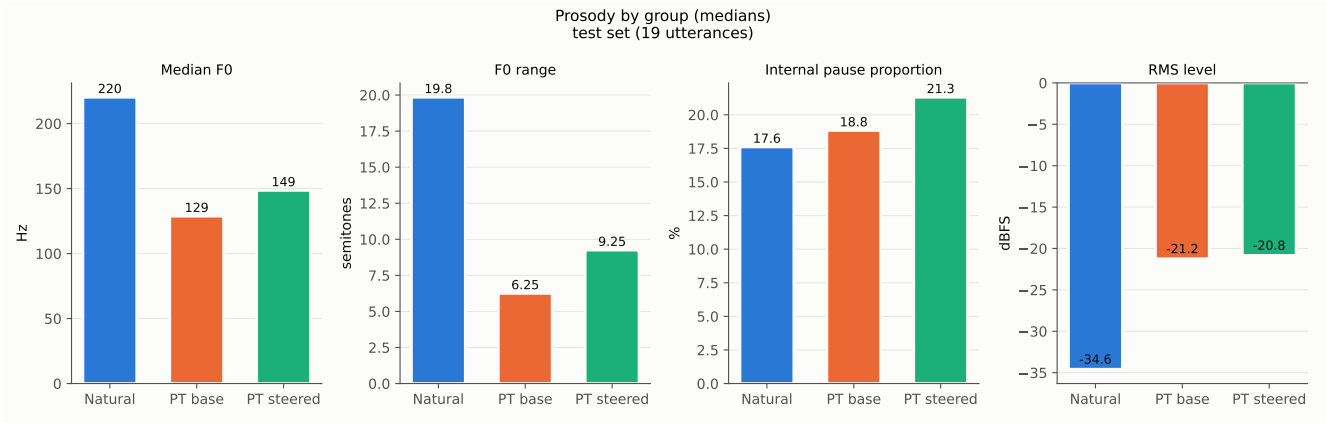


Figure 4: Prosody: median F0, F0 range, internal pause proportion, RMS level.

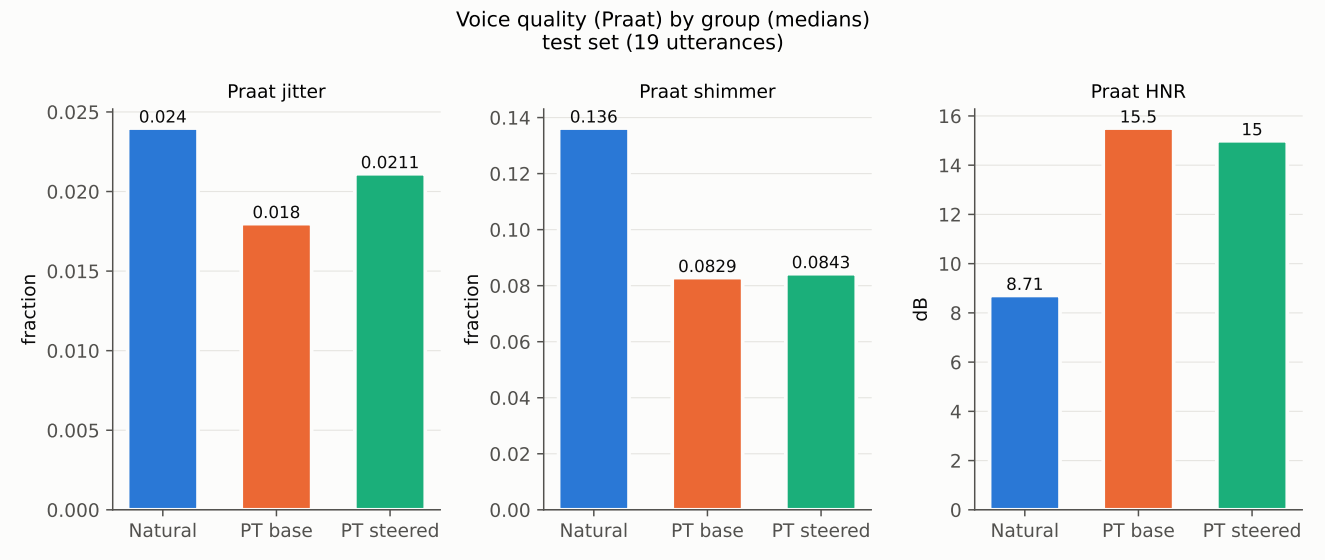


Figure 5: Voice quality: jitter, shimmer, HNR.

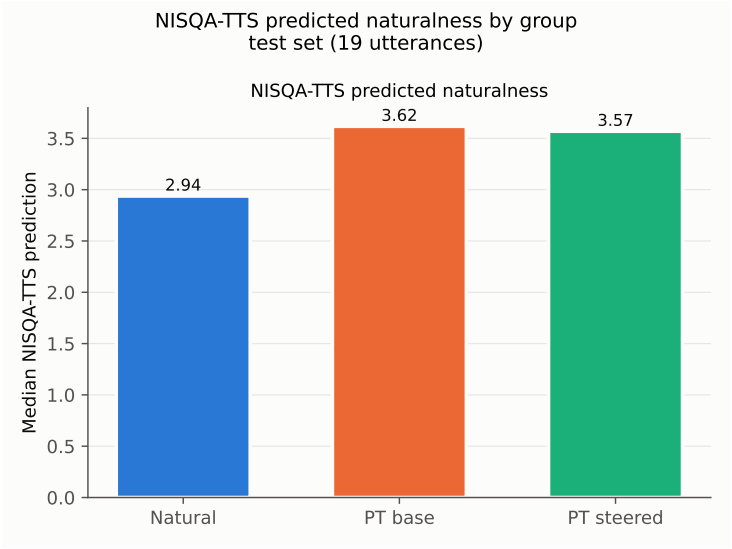


Figure 6: NISQA-TTS (learned quality score).

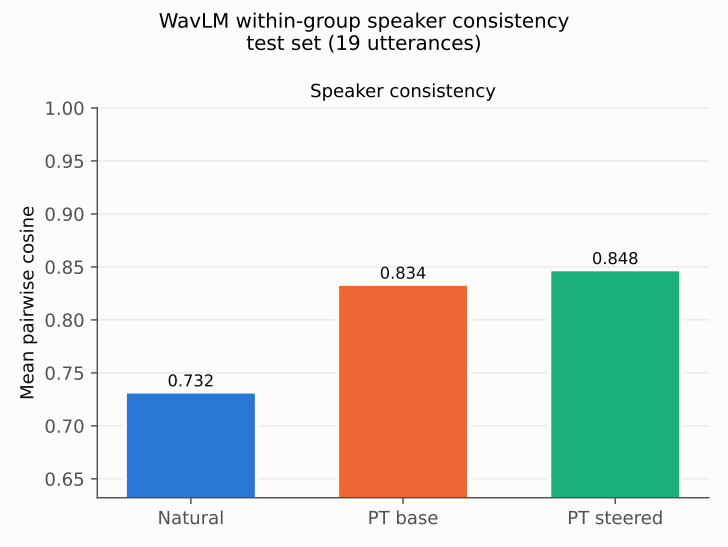


Figure 7: WavLM within-group speaker consistency (mean pairwise cosine). PT uses a random speaker per clip.

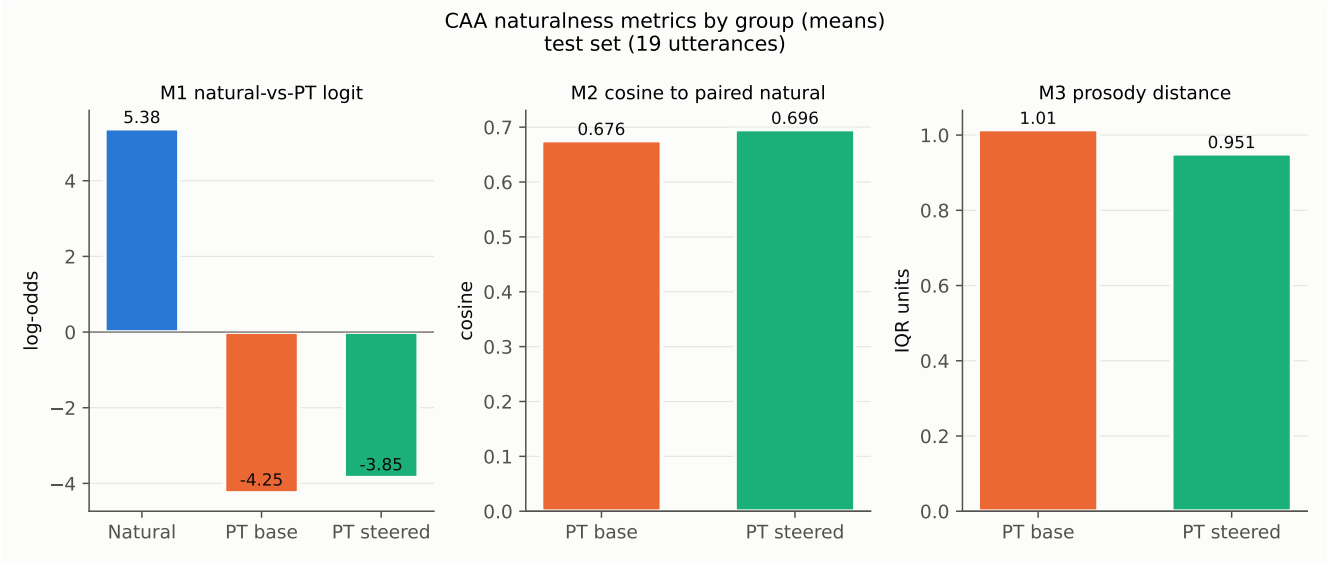


Figure 8: CAA metrics: M1 natural-vs-PT classifier logit, M2 cosine to paired natural clip, M3 prosody distance (lower is better).

2 All 100 utterances (in-sample)

The steering vector and the M1 classifier used the train utterances, so these numbers are biased in favour of steering.

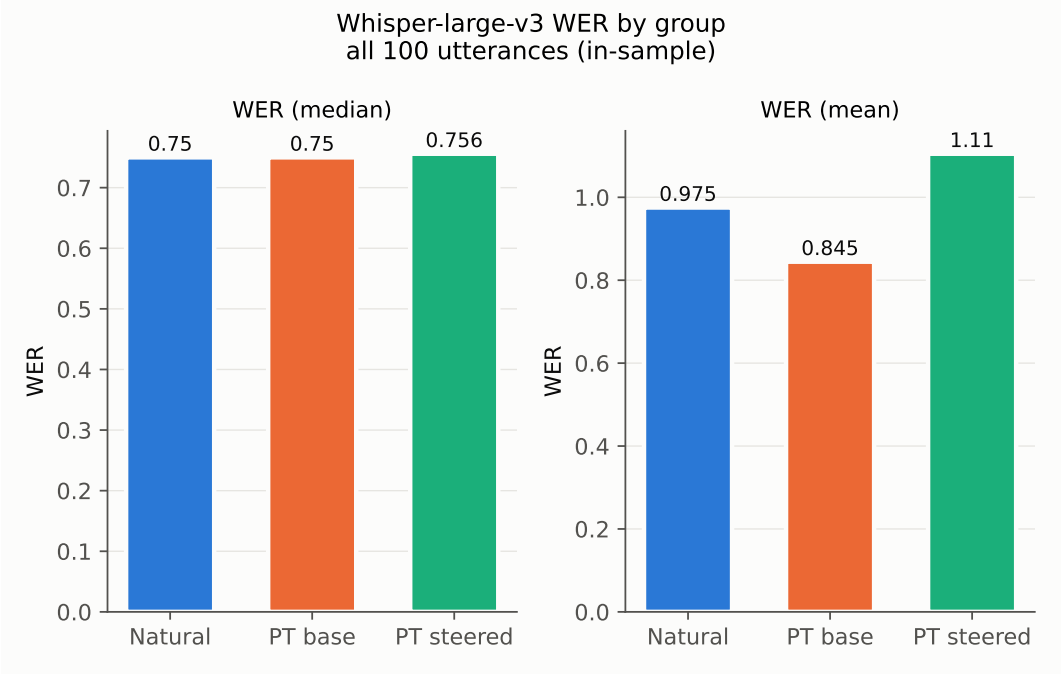


Figure 9: WER.

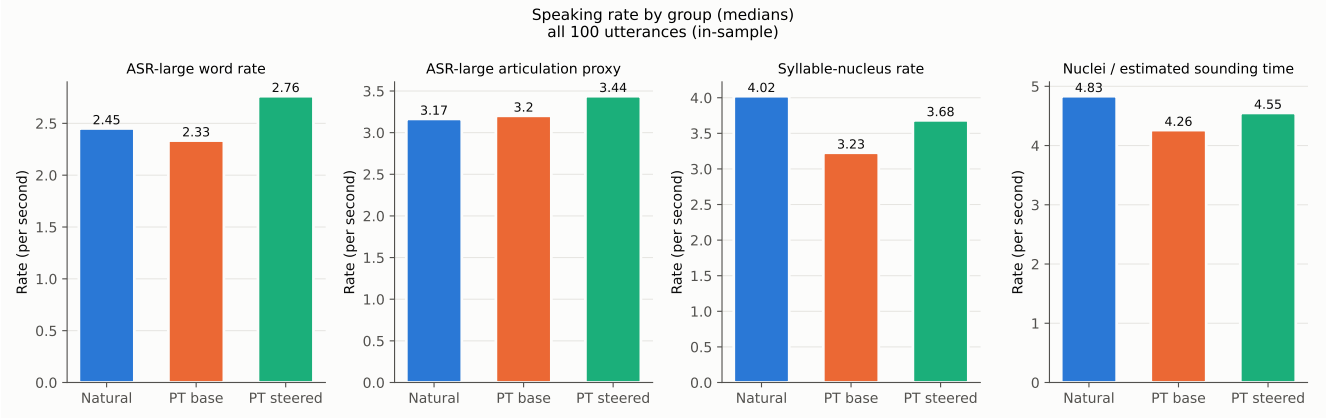


Figure 10: Speaking rate.

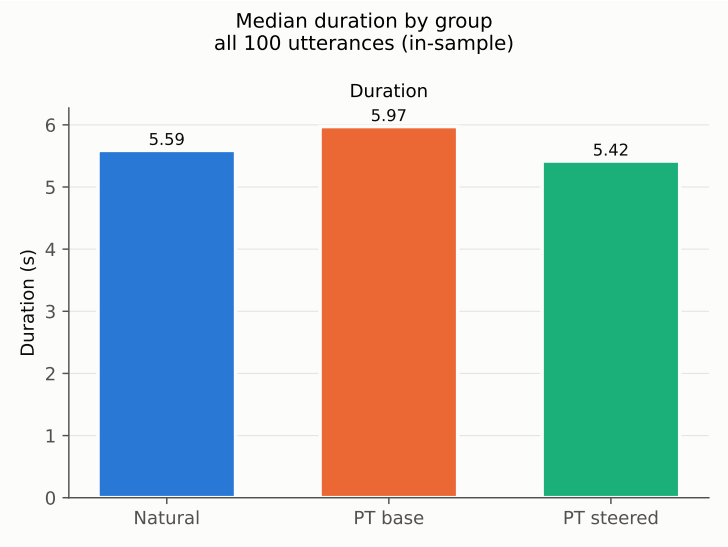


Figure 11: Median duration.

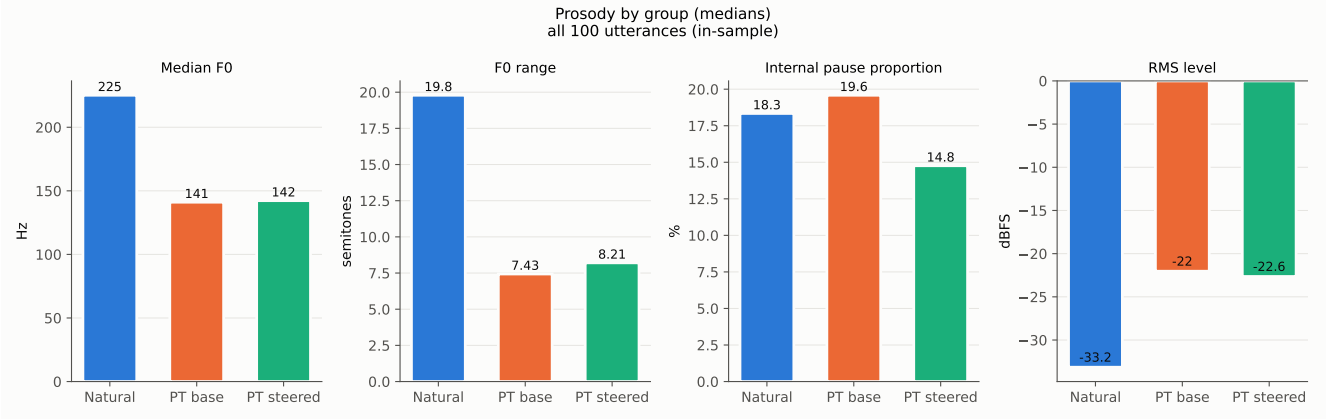


Figure 12: Prosody.

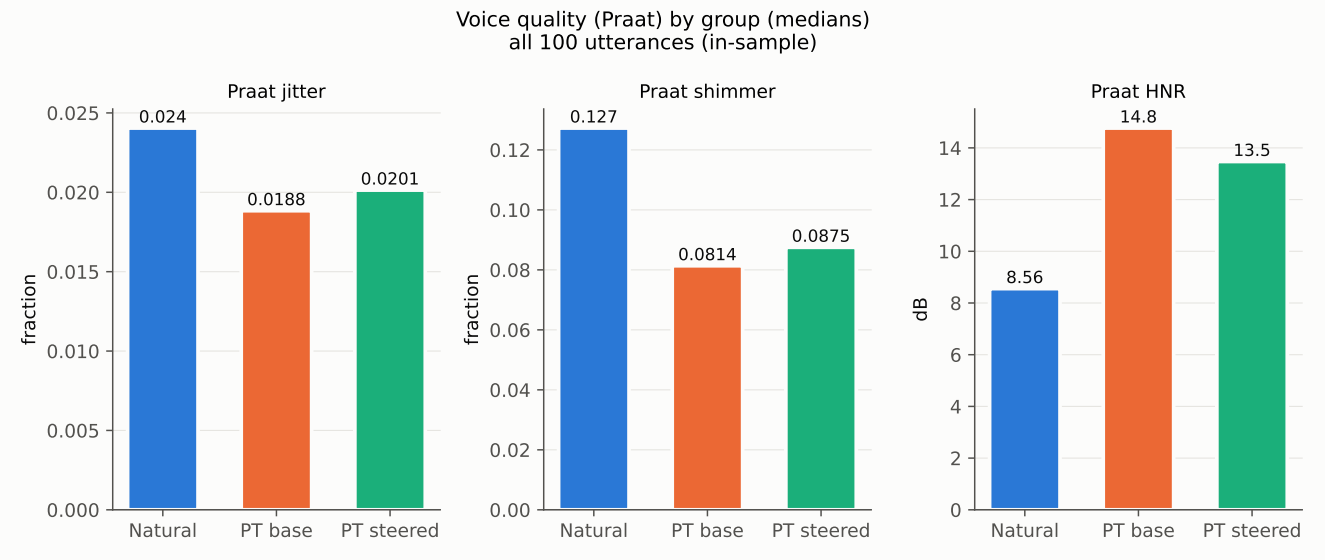


Figure 13: Voice quality.

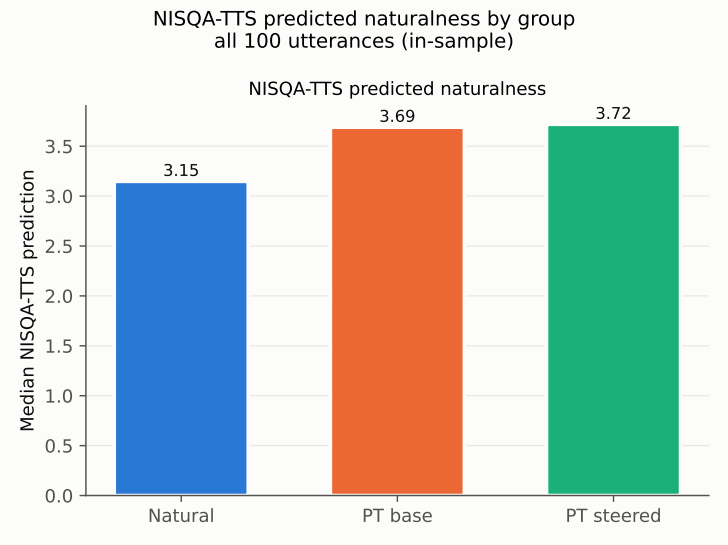


Figure 14: NISQA-TTS (one steered clip under 0.3 s unscored).

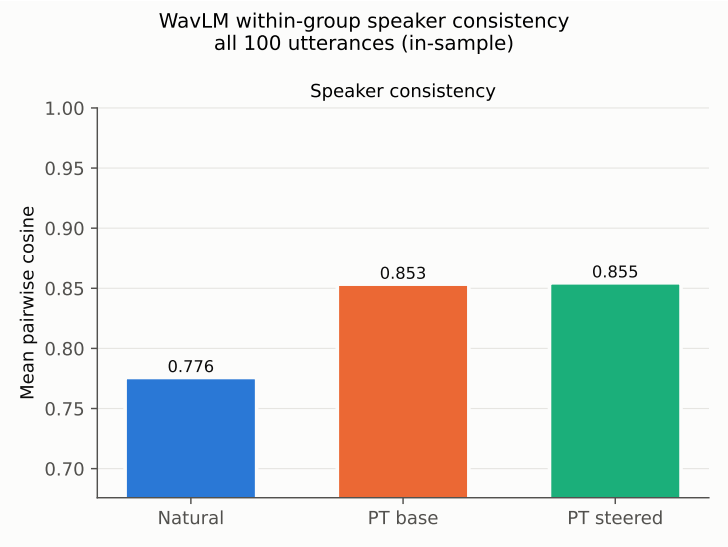


Figure 15: Speaker consistency.

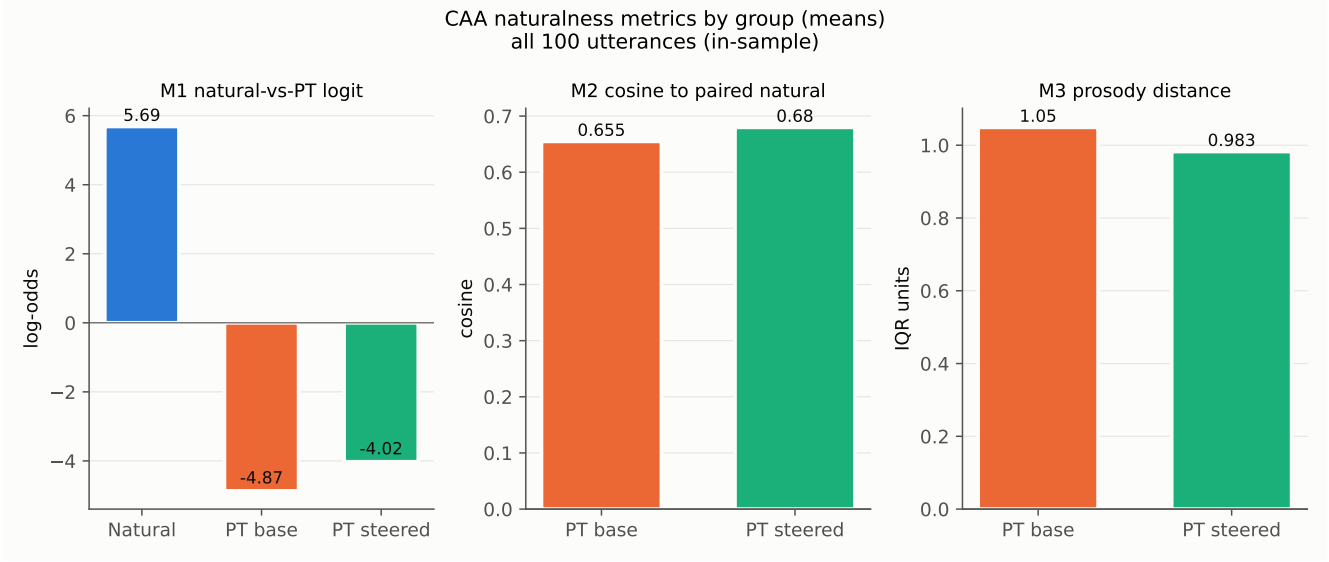


Figure 16: M1–M3.

Listening. `/local/speech/users/sp4050/Listening_Examples_100_PT/` on kedi: per utterance, the natural clip, `TRANSCRIPT.txt`, `02_OrpheusPT_base.wav`, `03_OrpheusPT_steered.wav`; open `listen.html`.